

United Nations General Assembly: Special Political and Decolonization Committee (UNGA-SPECPOL)

Study Guide

**AGENDA: Establishing legal barriers against the placement of
weapons in orbit and the terrestrial targeting of satellite
networks.**

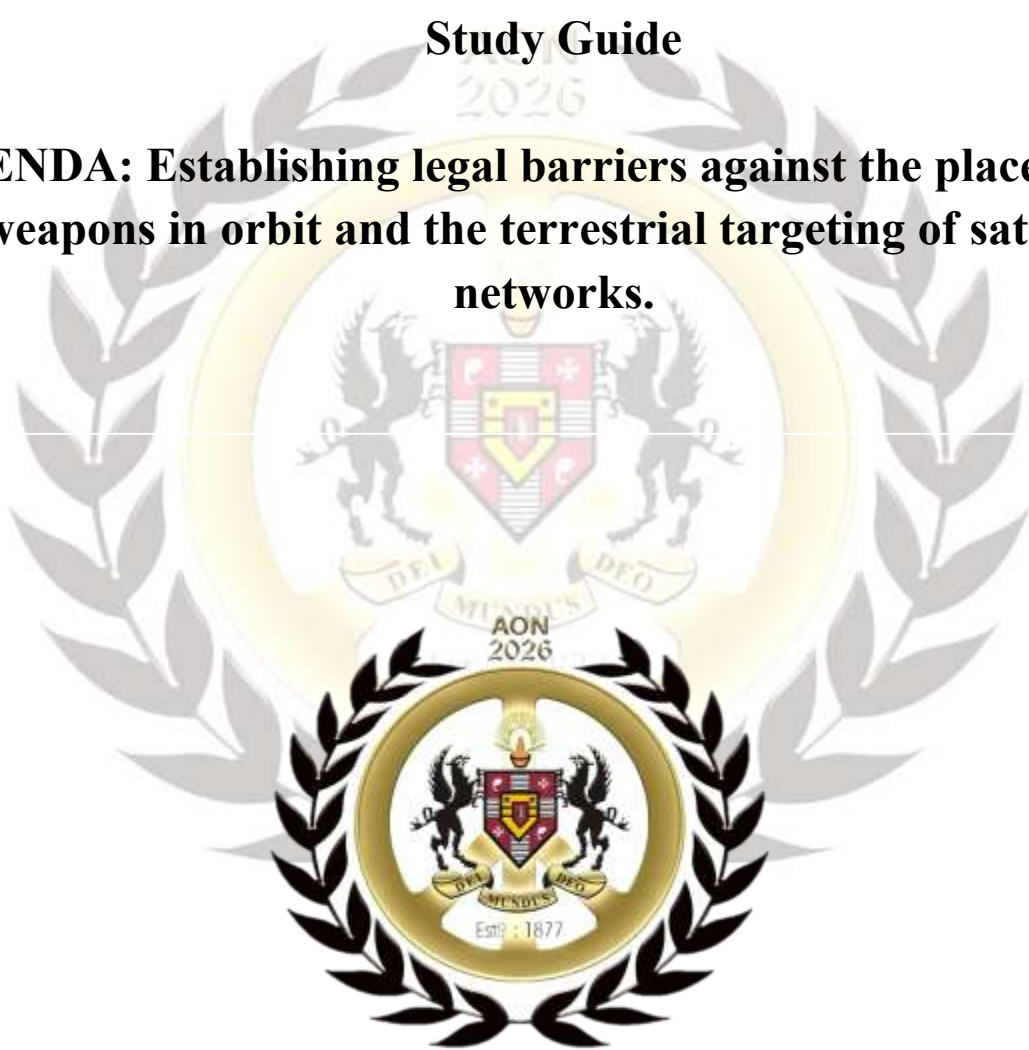


Table of Contents

1. Letter from the Executive Board
2. Flow of Debate Under Various Rules of Procedure
3. Standard of Proof
 - 3.1 Conclusive
 - 3.2 Rebuttable
 - 3.3 Application in Committee
4. Policy on Plagiarism and Use of Artificial Intelligence (AI)
5. Introduction to the United Nations General Assembly Special Political and Decolonization Committee (SPECPOL)
6. Mandate, Powers, and Functions of the Fourth Committee
 - 6.1 Decolonization and Self-Determination
 - 6.2 Peacekeeping and Special Political Missions
 - 6.3 Palestine Refugees and the Near East (UNRWA)
 - 6.4 International Cooperation in the Peaceful Uses of Outer Space
 - 6.5 Effects of Atomic Radiation
 - 6.6 Information and Communication Technologies for Peace
7. Agenda: Establishing Legal Barriers Against the Placement of Weapons in Orbit and the Terrestrial Targeting of Satellite Networks
 - 7.1 The Foundational Problem: From the ‘Peaceful Uses’ Principle to Modern Threats
 - 7.2 Core Definitions: What are ‘Weapons in Orbit’ and ‘Terrestrial Targeting of Satellites’?
 - 7.3 The Rise of Anti-Satellite (ASAT) Weapons: A Clear and Present Danger
 - 7.4 The Problem of ‘Dual-Use’ and ‘Space Basing’ of Conventional Weapons
 - 7.5 Terrestrial Targeting of Satellite Networks (The Ground Segment Threat)
 - 7.6 The Consequences of Weaponization: Debris, Escalation, and Global Public Goods
8. International Instruments and Soft Law Pertaining to the Agenda
 - 8.1 The Outer Space Treaty of 1967 and the ‘Province of All Mankind’

Principle

8.2 The Liability Convention of 1972: Fault, Forensics, and the Burden of Proof

8.3 The Registration Convention of 1975: Transparency Gaps

8.4 The Partial Test Ban Treaty of 1963 and Nuclear ASATs

8.5 The Anti-Ballistic Missile (ABM) Treaty of 1972

8.6 The Wassenaar Arrangement and MTCR: Export Controls as Technical Blockades

8.7 The Political Declaration on Responsible Behavior in Outer Space (2023–Present)

8.8 UNGA Resolutions: ‘No First Placement of Weapons in Outer Space’ (NFP)

8.9 The Draft Treaty on the Prevention of the Placement of Weapons in Outer Space (PPWT)

8.10 The International Code of Conduct for Outer Space Activities (EU-led Draft)

8.11 UN Group of Governmental Experts (GGE) and Open-Ended Working Group (OEWG) Reports

8.12 The 21 Guidelines for the Long-Term Sustainability (LTS) of Outer Space Activities

8.13 The Artemis Accords

9. Interesting and Unique Aspects of the Agenda

9.1 The ‘Space Basing’ Loophole

9.2 The Attribution Problem and Forced Blindness

9.3 The Verification Dilemma and the Impossibility of On-Orbit Inspections

9.4 The Enclosure of the Orbital Commons: Asymmetric Legal Frameworks and Privatized Warfare

9.5 ITAR, MTCR, and the Technical Blockade: Export Controls as a Barrier to Access

9.6 Non-Kinetic Interference, Directed-Energy Blinding, and Unannounced Rendezvous and Proximity Operations (RPOs) as Deniable Aggression

9.7 The Financial Weaponization of the Space Frontier: Insurance Asymmetries and Externalized Risk

9.8 The ‘Terrestrial Targeting’ Blind Spot and Ground Segment Vulnerability

- 10. Case Studies of Space Weaponization and Regulation Attempts
 - 10.1 The 2019 Indian ASAT Test: Responsible Testing or Dangerous Precedent?
 - 10.2 The Failure of the PPWT Negotiations (2008–2014): Lessons from a Collapsed Treaty
 - 10.3 The Amazon Rainforest Monitoring: Data Denial and Environmental Warfare
 - 10.4 The Space Insurance Market: Self-Insured Western Constellations and Systemic Premium Shocks for the Global South
 - 10.5 The Starfish Prime Test (1962): The Only Nuclear Detonation in Space
 - 11. Role of Other United Nations Bodies and External Actors
 - 12. Research Questions (Macro-Level & Position Shaping)
 - 13. Concluding Note – Resolution Expectations
-



1. LETTER FROM THE EXECUTIVE BOARD

From the Desk of Bureau,
Honorable Member State Representatives,

On behalf of the Executive Board (EB), it is our distinct privilege to welcome you to the ***United Nations General Assembly (UNGA) SPECPOL*** simulation at ***CBS AON 2026***. We look forward to a committee defined by rigorous analysis, principled diplomacy, and meaningful multilateral engagement.

The agenda for this session is:

“Establishing legal barriers against the placement of weapons in orbit and the terrestrial targeting of satellite networks.”

This agenda invites representatives to engage with one of the most significant and rapidly evolving challenges within contemporary international security and outer space governance. As reliance on satellite infrastructure continues to expand across civilian, commercial, scientific, and military domains, concerns regarding the weaponization of outer space and the vulnerability of space-based assets have become increasingly prominent within international discourse.

Representatives are expected to examine the legal, political, and strategic implications of placing weapons in Earth orbit, as well as the growing threat posed by terrestrial actions directed against satellite networks. Such actions may include anti-satellite (ASAT) weapons testing, electronic warfare capabilities, cyber operations targeting satellite systems, and other measures capable of disrupting or degrading critical space infrastructure. Particular attention should be given to the consequences of such activities for international peace and security,

global communications, navigation systems, disaster response mechanisms, scientific cooperation, and sustainable development.

Furthermore, representatives are encouraged to critically assess the adequacy of existing international legal frameworks governing outer space activities, including the principles established under the Outer Space Treaty and related instruments. Deliberations should explore existing legal gaps, challenges of verification and compliance, the balance between national security considerations and collective security interests, as well as the role of confidence-building and transparency measures in preventing the escalation of tensions in outer space.

Within the scope of this agenda, representatives are expected to ground their arguments in relevant General Assembly resolutions, reports of the Secretary-General, discussions conducted within the framework of the Open-Ended Working Group on reducing space threats through norms, rules and principles of responsible behaviours, the work of the Conference on Disarmament, and other applicable international legal and policy frameworks. Representatives should further consider the historical development of space law, contemporary state practice, and emerging technological capabilities that may affect the future security of outer space activities.

The Executive Board (EB) would like to place the following instructions on record:

1. Use of Study Guide:

Representatives are strongly advised to engage with the study guide in its entirety. However, the guide should not be regarded as an exhaustive source of information. Meaningful participation within committee will

require representatives to move beyond descriptive understanding and engage with the legal, political, technical, and strategic dimensions of the agenda. Independent research, critical evaluation of sources, and an understanding of national policy positions will be essential for productive deliberation.

2. Rules of Procedure (RoP):

This committee will function under the UN4MUN- integrated Indaba Rules of Procedure. Representatives unfamiliar with these rules are expected to acquaint themselves with the same prior to the conference. All representatives are required to maintain decorum, respect the authority of the Dais, and participate in proceedings with the professionalism expected of a United Nations General Assembly committee.

Prior to the commencement of the committee, representatives are encouraged to familiarize themselves with the institutional mandate of SPECPOL, particularly its responsibilities concerning international security, peaceful uses of outer space, disarmament-related discussions, and matters involving emerging political challenges before the international community. Delegates should also develop an understanding of existing space governance mechanisms, the positions of major space-faring nations, and the concerns of developing countries regarding equitable access to and peaceful use of outer space.

The Executive Board places particular emphasis on diplomacy that is both informed and solution-oriented. Representatives are encouraged to engage in constructive dialogue, build consensus wherever possible, and formulate recommendations that are legally sound, politically feasible, and consistent with the principles and purposes of the Charter of the

United Nations. Given the technical and strategic complexity of the agenda, delegates should strive to balance national interests with the broader objective of preserving outer space as a domain dedicated to peaceful purposes and the benefit of all humankind.

We look forward to an engaging and intellectually rigorous committee that reflects both the significance of the agenda and the responsibility entrusted to the Special Political and Decolonization Committee in addressing emerging challenges to international peace, security, and cooperation in outer space.

Warm regards,

Agnitra Saha

Chairperson

Adish Datta

Vice Chairperson

Chirag Chopra

Vice Chairperson

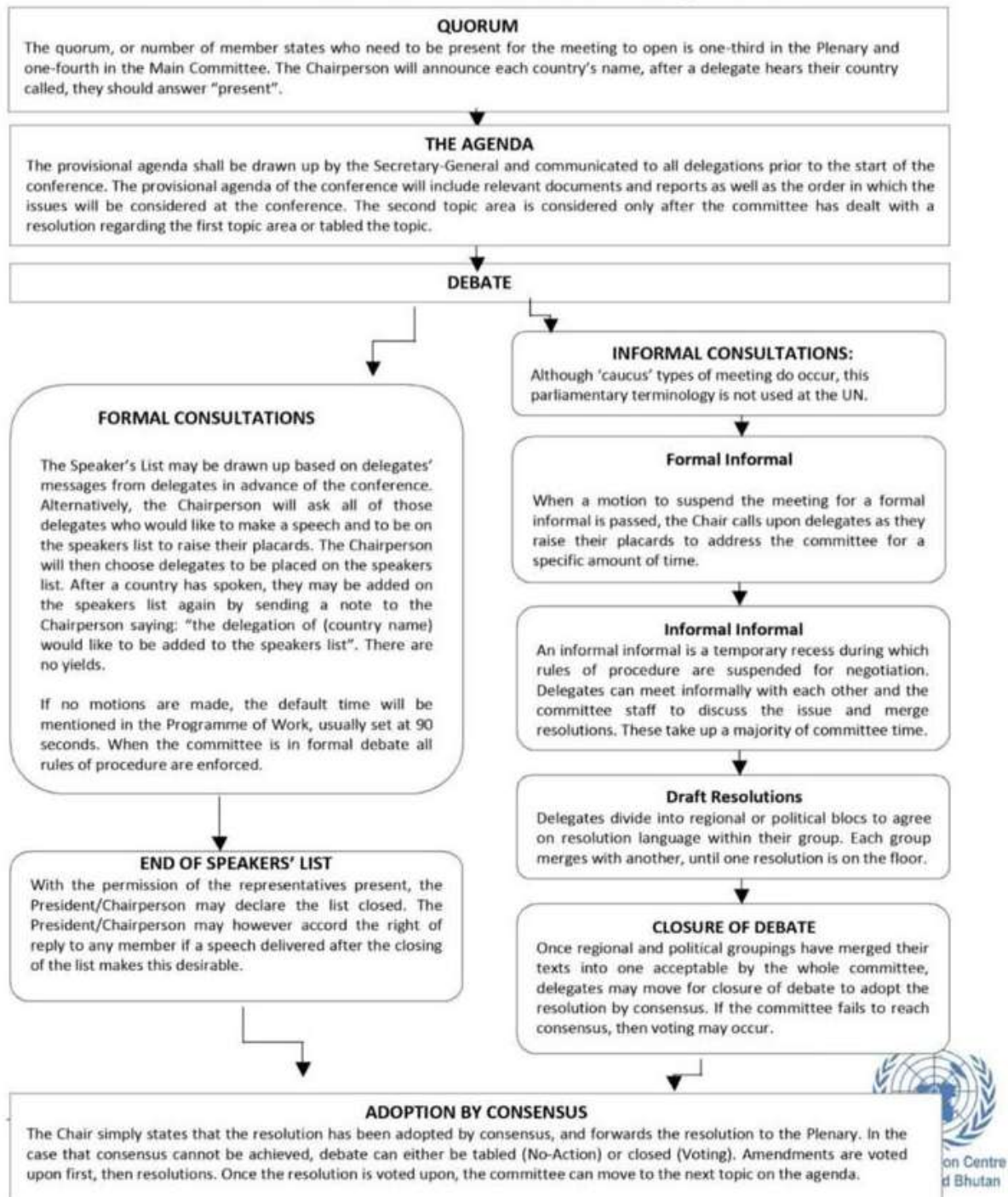
Executive Board, **United Nations General Assembly Special Political and Decolonization Committee (SPECPOL) | CBS AON 2026**

FLOW OF DEBATE UNDER VARIOUS RULES OF PROCEDURE-



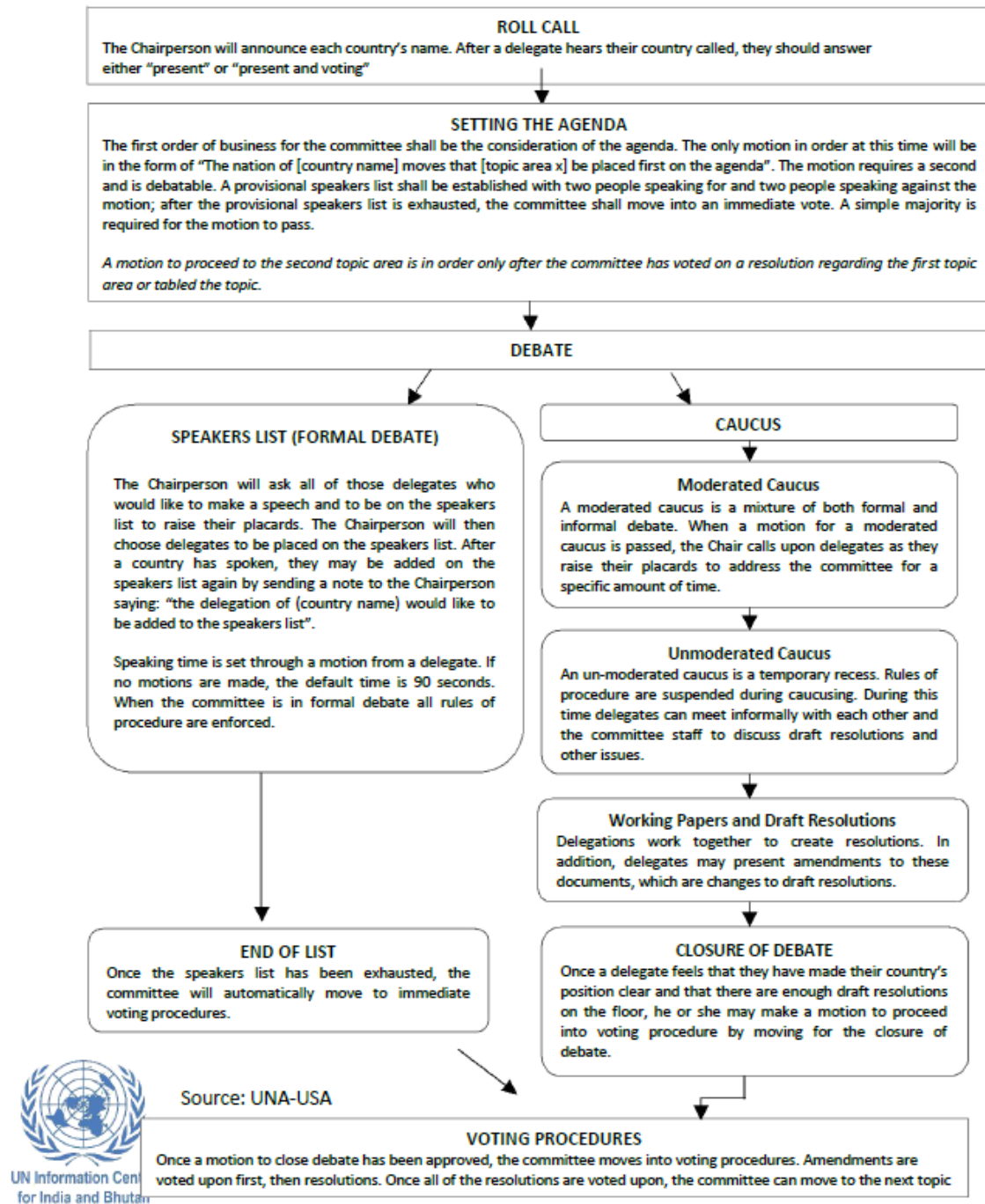
A. UN4MUN -

Flow of Debate: UN General Assembly Model



B. UNA-USA

Flow of Debate: Traditional MUN Model



C. INDABA

INDABA FORMAT OF DELIBERATION IN MUN

— A Consensus-Building Framework —



3. STANDARD OF PROOF

In order to ensure *procedural integrity* and *substantive accuracy* during deliberations, the Executive Board (EB) establishes the following standards with respect to the admissibility and evaluation of evidence presented in committee.

1. *All claims, allegations, and policy assertions made by Member State representatives* must be supported by credible and verifiable sources.

There shall broadly be *two* categories of proof that shall be submitted, namely:

- a) **Conclusive:** Conclusive Evidence refers to sources that are considered authoritative and carry a high degree of reliability, where these sources may be relied upon as factual unless directly contraindicated by equally credible material.
- b) **Rebuttable:** Rebuttable Evidence refers to sources that are informative but not inherently authoritative, where such evidence may be challenged, or superseded by more reliable or contradictory sources.

The following documents shall be acceptable as ‘proof’ of a claim/allegation in the course of the proceeding:

- **United Nations (UN) Reports & News (Conclusive)**, includes but not restricted to:
 - a) Reports of the United Nations (UN) Press;
 - b) Reports of the United Nations (UN) Organs and Agencies; and
 - c) Reports of the Allied Agencies i.e. considered authoritative (World Bank Group including IBRD, IDA etc. IMF, the GCC etc.)

● ***Government and its Agencies (Conclusive to the extent of controversy):*** Reports of any government and its agencies shall be conclusive so far as its own initiatives, schemes and achievements in a certain field are concerned. Similarly, any event or claim in controversy or remaining disputed as of the date of the session is rebuttable.

Example: A news report from a state-sponsored news agency shall not be conclusive proof of the absence of any state-sponsored human rights violations or persecution.

● ***New Sources (Rebuttable):***

a) Reports from Reuters, British Broadcasting Corporation (BBC) etc. shall only be accepted to the extent that there is no better proof on the contrary which disproves the claims in their articles. Similarly, publications from non-governmental organizations (NGOs), think tanks, and research institutions, unless supported by primary or official data, alongside secondary analyses, opinion pieces, and commentary-based sources can be used. A representative shall be allowed to use the article as a reference but is not given the guarantee that the claims in such an article shall be treated as proof.

Application in Committee:

a) Diplomatic Representatives are expected to clearly distinguish between verified facts and interpretative claims during formal debate.

b) Any serious allegation, particularly those concerning violations of international law, must be substantiated by conclusive evidence.

c) Rebuttal evidence may be used to support arguments, however, it remains open to challenge and may not be treated as definitive proof.

d) *The Executive Board (EB) reserves the discretion* to question, accept, or dismiss evidence presented in committee if it does not meet the required standards of credibility and relevance.

Note: This framework is intended to promote fact-based deliberation, responsible diplomacy, and analytical rigour, in line with the expectations of a United Nations General Assembly SPECPOL.

4. POLICY ON PLAGIARISM AND USE OF ARTIFICIAL INTELLIGENCE (AI)

All forms of documents, including, but not limited to, Draft Resolutions (DR), Position Papers, Press Releases etc., would be checked for plagiarism and use of Artificial Intelligence (AI). Bearing in mind the norms and formatting requirements prescribed for United Nations (UN) documentation, the extent of plagiarism would be capped at 15%. Further, as a general rule, the use of Artificial Intelligence (AI) would not be tolerated at any juncture during the proceedings, but given the futile and primitive nature of most free-use software for the detection of Artificial Intelligence (AI), the Dais shall be setting the cap at 5%.

5. INTRODUCTION TO THE UNITED NATIONS GENERAL ASSEMBLY SPECIAL POLITICAL AND DECOLONIZATION COMMITTEE (SPECPOL)

The United Nations General Assembly Special Political and Decolonization Committee, commonly known as the Fourth Committee or SPECPOL, constitutes one of the six main committees of the General Assembly. It was established to handle a specific set of agenda items that did not fit neatly into other committees, primarily those related to decolonization, the effects of atomic radiation, and unique political questions requiring specialized attention. Over time, its mandate has expanded to include international cooperation in the peaceful uses of outer space, peacekeeping operations, and information cooperation. Unlike the First Committee (Disarmament and International Security), which deals with nuclear weapons and conventional arms control, or the Sixth Committee (Legal), which deals with the progressive development of international law, the Fourth Committee operates in a space where political consensus, technical expertise, and humanitarian outcomes must be balanced. It is a committee where the distinction between a "security" issue and a "political" issue is often blurred.

Within the institutional architecture of the United Nations, the Fourth Committee functions as a forum where Member States collectively deliberate on issues that are political in nature but require technical expertise and broad international cooperation. Unlike the Security Council, which possesses binding enforcement authority under Chapter VII of the Charter, the General Assembly and its committees primarily exercise recommendation authority, shaping international norms, guiding institutional practice, and facilitating consensus on emerging global challenges. Resolutions passed by SPECPOL are recommendations to the General Assembly plenary and, if

adopted there, become non-binding but politically powerful expressions of the international community's will. The Assembly's involvement is grounded in its broader mandate under the Charter of the United Nations to discuss and make recommendations on matters relating to international peace and security.

The Fourth Committee is distinct from the First Committee (Disarmament and International Security) in a manner that is crucial for your agenda. The First Committee deals with nuclear weapons, conventional arms, and other security threats in a direct disarmament context.

SPECPOL approaches space security from the perspective of preventing an arms race and ensuring space remains a global public good for all nations, including developing countries that rely on space-based services for agriculture, disaster management, and communication. This distinction means that your resolution should not be framed purely as a disarmament text. It must also address the positive uses of space, capacity-building for developing states, and the protection of civilian infrastructure. Your task is to uphold this mandate.

6. MANDATE, POWERS, AND FUNCTIONS OF THE FOURTH COMMITTEE

The mandate and authority of the Fourth Committee derive primarily from the rules of procedure of the General Assembly and specific agenda items assigned to it by the plenary. Unlike the Security Council, which possesses binding enforcement authority under Chapter VII of the

Charter, the General Assembly and its committees primarily exercise recommendation authority, shaping international norms, guiding institutional practice, and facilitating consensus on emerging global challenges. The Fourth Committee exercises these functions across several distinct areas.

a) Decolonization and Self-Determination

This is the historical core mandate of the Fourth Committee. The committee examines the implementation of the Declaration on the Granting of Independence to Colonial Countries and Peoples and hears petitioners from non-self-governing territories. The committee reviews the status of the 17 remaining non-self-governing territories and makes recommendations on their future. While not directly relevant to space weapons, this function establishes the committee's expertise in managing contested political statuses and the principle of self-determination, which has parallels in debates about sovereignty over celestial body resources.

b) Peacekeeping and Special Political Missions

The Fourth Committee reviews the comprehensive reports on peacekeeping operations, including their financing, mandates, and administrative arrangements. The committee examines the budgets of all active peacekeeping missions and special political missions, which currently number over a dozen. This is highly relevant because space-based assets, including communications satellites, surveillance satellites, and navigation systems, are critical for modern peacekeeping. An attack on a satellite network used by a UN peacekeeping mission could

endanger the lives of peacekeepers and compromise the success of the mission. The committee's oversight function allows it to request reports on the vulnerability of peacekeeping communications to space-based threats.

c) Palestine Refugees and the Near East (UNRWA)

The committee reviews the work of the United Nations Relief and Works Agency for Palestine Refugees. This is a specific, recurring agenda item that is not directly relevant to space weapons, though it demonstrates the committee's role in addressing protracted humanitarian situations and its ability to manage complex, politically sensitive mandates alongside technical ones.

d) International Cooperation in the Peaceful Uses of Outer Space

This is your core mandate. Under the agenda item "International cooperation in the peaceful uses of outer space," the Fourth Committee reviews the annual report of the Committee on the Peaceful Uses of Outer Space (COPUOS). The committee promotes the benefits of space technology for all nations, encourages capacity-building, and works to prevent an arms race in outer space (PAROS). This is where your agenda resides. The committee does not negotiate treaties, but it provides the political direction for COPUOS and other bodies. The Fourth Committee's function here is to translate the technical work of COPUOS into political recommendations for the General Assembly.

The committee's work in this area is guided by several key principles derived from the Outer Space Treaty. First, the exploration and use of outer space shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development. Second, outer space is not subject to national appropriation by claim of sovereignty. Third, states parties to the Treaty undertake not to place in orbit around the Earth any objects carrying nuclear weapons or any other kinds of weapons of mass destruction. The Fourth Committee's mandate is to elaborate on these principles and to extend them to address conventional weapons and terrestrial threats not covered by the original treaty.

e) Effects of Atomic Radiation

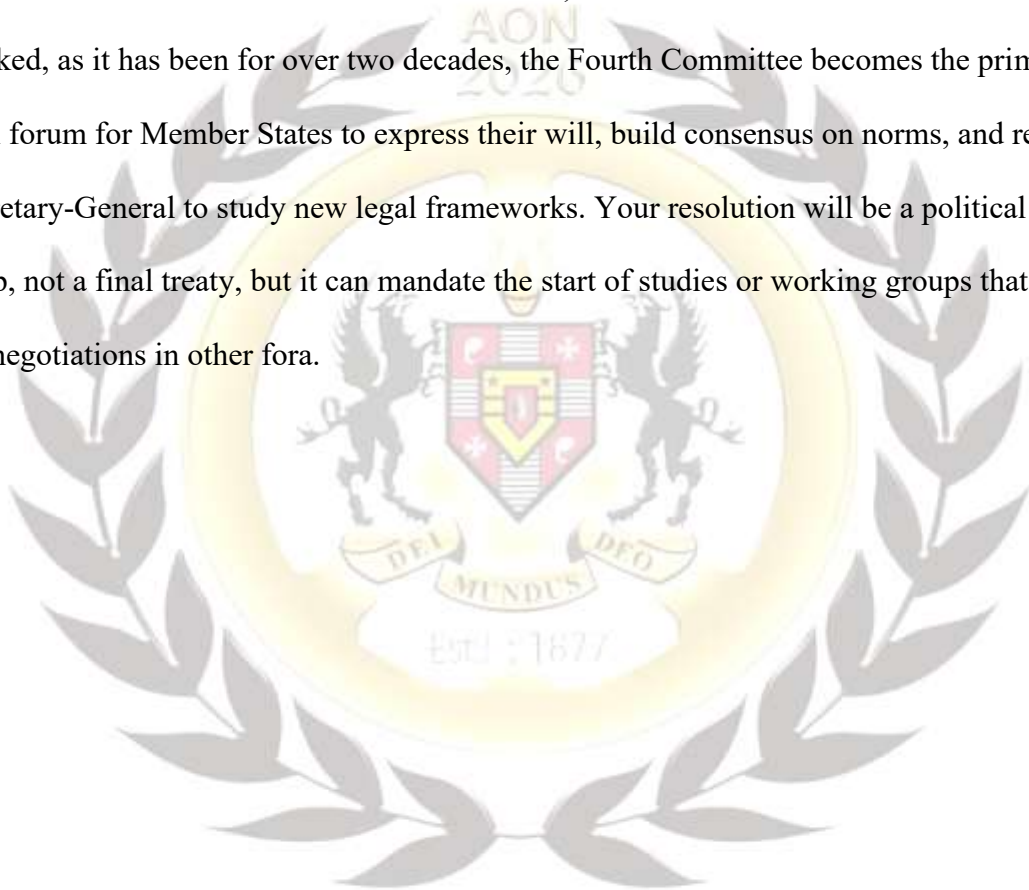
The committee reviews the work of the United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR). While not directly relevant to space weapons, it is worth noting that nuclear detonations in space are banned under the Partial Test Ban Treaty, and the effects of space-based radiation on electronics are relevant to satellite vulnerability. The committee's work on radiation effects provides a scientific basis for assessing the environmental consequences of a nuclear detonation in space, which would be a catastrophic event for satellite networks.

f) Information and Communication Technologies for Peace

A newer area of focus, this involves using technology to prevent conflict and build trust. This connects directly to the cyber threats against satellite ground stations and the use of confidence-building measures to prevent misinterpretation of space activities. The committee has

increasingly recognized that information and communication technologies, including those used for satellite command and control, are dual-use and require governance frameworks that balance security and access.

Crucially, the Fourth Committee is not a treaty-negotiating body. The Conference on Disarmament in Geneva has that mandate. However, when the Conference on Disarmament is deadlocked, as it has been for over two decades, the Fourth Committee becomes the primary political forum for Member States to express their will, build consensus on norms, and request the Secretary-General to study new legal frameworks. Your resolution will be a political roadmap, not a final treaty, but it can mandate the start of studies or working groups that could lead to negotiations in other fora.



7. AGENDA: ESTABLISHING LEGAL BARRIERS AGAINST THE PLACEMENT OF WEAPONS IN ORBIT AND THE TERRESTRIAL TARGETING OF SATELLITE NETWORKS

a) The Foundational Problem: From the ‘Peaceful Uses’ Principle to Modern Threats

The 1967 Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies (Outer Space Treaty), is the foundational instrument of international space law. It was drafted during the Cold War, a period of intense geopolitical rivalry between the United States and the Soviet Union. The drafters were primarily concerned with two scenarios: the placement of nuclear weapons in orbit and the extension of territorial sovereignty to the Moon and other celestial bodies. These concerns reflected the immediate fears of the era: that the Soviet Union would place nuclear-armed satellites in orbit, creating a permanent and unstrikeable first-strike capability, or that either superpower would claim sovereignty over parts of the Moon as an extension of terrestrial territorial disputes.

Article IV of the Outer Space Treaty addresses these concerns directly: "States Parties to the Treaty undertake not to place in orbit around the Earth any objects carrying nuclear weapons or any other kinds of weapons of mass destruction, install such weapons on celestial bodies, or station such weapons in outer space in any other manner." The Moon and other celestial bodies

are to be used "exclusively for peaceful purposes." The establishment of military bases, the testing of weapons, and the conduct of military maneuvers on celestial bodies are forbidden.

This was a remarkable achievement for its time. It banned the most catastrophic weapons from orbit. It established the principle that outer space should be used for peaceful purposes. It created a legal foundation that has served the international community for over five decades. The treaty has 116 states parties, including all major spacefaring nations (the United States, Russia, China, India, Japan, and the member states of the European Space Agency). It is widely considered to have achieved the status of customary international law, meaning that even non-parties are bound by its core principles.

However, the treaty was a product of its technological era. The drafters could not foresee the technological developments of the subsequent decades. In 1967, placing a conventional weapon in orbit was impractical. The technology did not exist to make it accurate or cost-effective. The drafters could not foresee the miniaturization of electronics, the development of precision-guided munitions, the invention of the laser, the emergence of cyber-weapons, or the commercialization of space. The treaty's prohibition is limited to weapons of mass destruction. Conventional weapons, including kinetic interceptors, lasers, and cyber-tools, are not banned. A state could legally place a conventional missile interceptor in orbit today, as long as it does not carry a nuclear warhead.

Furthermore, the Outer Space Treaty did not envision the terrestrial targeting of satellite networks. The drafters assumed threats to satellites would come from other objects in space.

They did not consider that a state could launch a cyber-weapon from a laptop to disable a satellite ground station, or fire a missile from a fighter jet to destroy a ground antenna. The treaty has no provisions addressing attacks on the ground infrastructure that controls satellites. The concept of a "cyber-attack" did not exist in 1967. The International Telecommunication Union (ITU) regulated radio interference, but the idea of a "denial of service" attack on a ground station's network was science fiction.

Therefore, the modern problem has two interconnected parts. First, weapons placed in orbit that can attack other satellites or targets on Earth. Second, attacks originating from Earth against satellite networks, including ground stations, data links, and the satellites themselves via electronic or cyber means. Any comprehensive legal barrier must address both. The central challenge is to extend the legal framework of the Outer Space Treaty to cover technologies and threats that its drafters could not have imagined, without reopening the entire treaty to renegotiation. Renegotiation carries the risk that states would seek to weaken existing provisions or introduce new exceptions. The existing treaty, despite its gaps, remains the bedrock of international space law. The approach taken by the international community has therefore been to supplement the treaty with additional instruments, codes of conduct, and transparency and confidence-building measures, rather than to rewrite it.

b) Core Definitions: What are ‘Weapons in Orbit’ and ‘Terrestrial Targeting of Satellites’?

Before any legal barrier can be established, the terms must be defined. There is currently no universally agreed definition of a space weapon. This is one of the primary roadblocks to progress, as states fear that a definition that is too broad could restrict legitimate peaceful activities, while a definition that is too narrow would create loopholes. The following definitions are drawn from the scholarly literature, including the Woomera Manual on the International Law of Military Space Operations, the MILAMOS Manual, and the work of the UN Group of Governmental Experts.

Weapon in Orbit: For the purpose of international legal analysis, a weapon in orbit can be defined as a space object, or a component of a space object, placed in Earth orbit or beyond, whose primary or secondary design purpose is to damage, destroy, disable, or otherwise interfere with the normal functioning of another space object, a terrestrial object, or human life. This definition includes both the object itself and any component that can be used as a weapon. It includes several categories, each with distinct technical characteristics and legal challenges.

Kinetic interceptors are weapons that physically collide with a target. A kinetic interceptor can be a dedicated "kill vehicle" or a satellite that uses its own mass as a projectile. The damage is caused by the force of impact at extremely high speeds. In low Earth orbit, orbital velocities are approximately 7.8 kilometers per second (17,500 miles per hour). A kinetic impact at these

velocities releases energy equivalent to several kilograms of high explosive for every kilogram of interceptor mass. A 1 kilogram object impacting at 7.8 km/s releases approximately 30 megajoules of kinetic energy, roughly equivalent to 7 kilograms of TNT. The destructive potential is enormous. Kinetic interceptors can be launched from Earth (direct-ascent) or from another satellite in orbit (co-orbital). Direct-ascent kinetic ASATs are missiles launched from ground, air, or sea platforms. They are detectable by missile warning systems. Co-orbital kinetic ASATs are satellites placed in orbit that maneuver to intercept a target. They are harder to detect because they can be launched as routine satellite launches and can loiter for months or years before attacking.

Directed-energy weapons include lasers and high-power microwaves. A laser can permanently blind the sensitive optical sensors on a satellite, or it can physically damage components through thermal heating. A high-power microwave weapon can overwhelm or destroy a satellite's electronics without any physical impact. These weapons are particularly difficult to attribute because they leave no debris and can be operated from ground, air, or space. A ground-based laser used for astronomical observation can be repurposed as a weapon. The act of "dazzling" a satellite (temporarily blinding it) sits in a legal grey zone: it is not permanent destruction, but it constitutes a denial of service that can have military and civilian consequences. The 1997 incident in which a Russian ground-based laser allegedly dazzled a Japanese Earth observation satellite remains unresolved. Space-based lasers would be weapons placed in orbit.

Distinguishing between a peaceful LIDAR system (used for earth observation) and a weaponized laser is extremely difficult based on external observation alone. Both require a laser emitter. The only difference is the power level and the targeting system.

c) The Rise of Anti-Satellite Weapons (ASATs): Refer [UN General Assembly Resolution 77/41 \(2022\)](#),

Please understand the connotations and impact of these terminologies:

1. Direct-Ascent Kinetic ASATs
2. Co- Orbital ASATs
3. Non Kinetic ASATs
4. Nuclear Armed ASATs

d) [The Problem of 'Dual-Use' and 'Space Basing' of Conventional](#)

[Weapons](#)- *The document explains why distinguishing between dual-use and dual-purpose space systems is important for improving space security and regulation.*

[\(Also refer to this second highlight\)](#)- *The document explains how international law, especially the Law of the Sea and Space Law, can guide responsible state behaviour in cyberspace.*

e) Terrestrial Targeting of Satellite Networks

(The Ground Segment Threat)

This aspect of the agenda is often overlooked but is strategically critical. A satellite is a vulnerable object in a harsh environment. However, its ground station is even more vulnerable.

The ground station is a building on Earth with antennas, computers, and people. It can be bombed, sabotaged, or hacked. An attacker who cannot reach the satellite itself can attack the ground station that controls it. This is often easier and cheaper than attacking the satellite.

Cyber-Attacks on Ground Stations

Satellites are controlled from ground stations. An attacker who gains access to the ground station's computer network can send commands to the satellite. The attacker can change the satellite's orbit, turn off its systems, or cause it to transmit incorrect data. In 2018, a United States government report noted that foreign hackers had gained access to the ground stations of NOAA weather satellites. The hackers had obtained credentials and could have taken control of the satellites. The report did not attribute the attack to any specific state.

The legal classification of cyber-attacks on ground stations is uncertain. Under the UN Charter, a use of force is prohibited. A cyber-attack that causes physical damage to a satellite (e.g., overwriting firmware causing a satellite to become unusable) is a use of force. However, a cyber-attack that merely disrupts service without causing physical damage (e.g., a denial of service attack that prevents commands from being sent) is in a grey zone. The implication for satellite ground stations is that a cyber-attack that takes a ground station offline, preventing commands from being sent to a satellite, may not be legally considered a use of force, even though the effect on the satellite is the same as a physical attack.

Electronic Warfare: Jamming and Spoofing

Jamming is the transmission of radio signals on the same frequency used by a satellite, overwhelming the signal and blocking communication. Jamming can be conducted from a ground-based transmitter, from an aircraft, or from another satellite. It is a common tactic in conflicts. In Ukraine, both Russian and Ukrainian forces have used jamming to disrupt each other's communications and drone operations. Jamming does not physically damage the satellite, but it denies the user access to the satellite's services.

Spoofing is more sophisticated. The attacker transmits a false signal that mimics the real satellite signal, tricking the receiver. A GPS spoofer can cause a receiver to calculate the wrong position. In 2011, a US drone was captured by Iranian forces; it is believed that Iranian forces spoofed the GPS signal, causing the drone to land at the Iranian site rather than returning to its base. Spoofing is particularly dangerous because the victim may not know that the signal is false. A ship's navigation system will show a position that appears correct, but it is a fabrication.

The International Telecommunication Union (ITU) has rules against "harmful interference" with radio services. The ITU Constitution defines harmful interference as "interference which endangers the functioning of a radionavigation service or of other safety services or seriously degrades, obstructs or repeatedly interrupts a radiocommunication service operating in accordance with these Regulations." However, the ITU's rules apply to civilian telecommunications. Military jamming during an armed conflict is not clearly prohibited. The ITU has no enforcement mechanism against states that engage in military jamming.

Physical Attacks on Ground Stations

A cruise missile strike on a ground station is clearly an act of armed force under the UN Charter. It would trigger the right of self-defence under Article 51. However, the Outer Space Treaty does not specifically address attacks on ground stations. The Liability Convention applies to damage caused by a space object, not damage caused to a space object. An attack on a ground station is an attack on terrestrial infrastructure, governed by the laws of armed conflict (IHL), not space law. Under IHL, a ground station that is used for military purposes (e.g., commanding a military satellite) is a military objective and can be attacked. A ground station that is used exclusively for civilian purposes (e.g., broadcasting television) is a civilian object and cannot be attacked. However, many ground stations are dual-use. The same ground station that commands a military communications satellite also commands a civilian communications satellite. Under IHL, if a ground station makes an effective contribution to military action, it is a military objective. The fact that it also serves civilian purposes does not immunize it, but the principle of proportionality requires that the incidental harm to civilians not be excessive. This is a high bar, but it is not an absolute prohibition.

f) The Consequences of Weaponization: Debris, Escalation, and Global Public Goods

The consequences of space weaponization are global and affect every person on Earth. The following analysis draws on the legal scholarship on self-defence in space, indiscriminate attacks, and the laws of armed conflict.

The Kessler Syndrome

Named after NASA scientist Donald Kessler, this is a scenario in which the density of objects in Low Earth Orbit becomes high enough that a single collision creates enough debris to cause further collisions, creating a cascade. Once the cascade begins, it continues until the orbit is filled with a cloud of debris. Low Earth Orbit would become unusable for decades or centuries. Weather forecasting would degrade. Satellite communications would end. GPS and other navigation systems would fail. Earth observation for agriculture, disaster management, and climate monitoring would stop. The Kessler Syndrome is not science fiction. It is a plausible outcome of continued ASAT testing and space weaponization. A 2011 report by the National Research Council concluded that the debris environment in low Earth orbit has already reached a "tipping point" where collisions will continue to increase even without new launches.

From a legal perspective, the Kessler Syndrome engages Article IX of the Outer Space Treaty, which requires states to avoid harmful contamination of space and to consult if an activity could cause harmful interference. It also engages Articles 35 and 55 of Additional Protocol I to the Geneva Conventions (1977). Article 35(3) prohibits the employment of methods or means of warfare which are intended, or may be expected, to cause widespread, long-term and severe damage to the natural environment. Article 55(1) requires care to be taken to protect the natural environment against such damage. The commentary to Additional Protocol I states that the natural environment should be understood in the widest sense, including outer space. A debris cascade is "widespread" (it affects the entire orbital band), "long-term" (debris persists for

decades or centuries), and "severe" (it destroys the utility of the orbit). The use of kinetic ASAT weapons would almost certainly violate these provisions.

The Escalation Ladder from Space to Nuclear Conflict

Nuclear-armed states rely on early-warning satellites to detect missile launches. These satellites are stationed in geostationary orbit (approximately 35,786 kilometers altitude) and constantly watch for the infrared signature of a ballistic missile launch. The US Space-Based Infrared System (SBIRS) and the Russian Oko system are examples. If a state were to attack another state's early-warning satellites, it could blind that state to an incoming missile attack. The victim state would face an impossible decision: launch its missiles on warning (before it has confirmed an attack) or risk being destroyed by a first strike that its blinded satellites did not detect.

Space weaponization dramatically increases the risk of accidental nuclear war. A non-kinetic ASAT attack that does not destroy a satellite but temporarily blinds it could be misinterpreted as the first step of a nuclear attack. A cyber-attack that causes a satellite to send false data could create a false warning of a missile launch. The present agenda is therefore not only about space security but about nuclear stability. The *jus ad bellum* requirements of necessity and proportionality apply to ASAT weapon use. A state may only use force in self-defence if an armed attack is imminent and if the response is proportionate. An attack on an early-warning satellite would almost certainly meet the armed attack threshold, because it would constitute an attack on a state's ability to detect a nuclear first strike. However, a proportionate response might be limited to disabling the attacking satellite, not launching a nuclear strike.

The Threat to Civilian Infrastructure

Modern civilian infrastructure is entirely dependent on satellite networks. GPS is used for timestamping financial transactions. When a credit card or ATM transaction occurs, the transaction time is synchronized by GPS. Power grids use GPS timing to synchronize alternating current. If GPS timing fails, power grids can become unstable and shut down. The 2013 cyber-attack on the US power grid, which was not GPS-related, demonstrated the vulnerability of power infrastructure. A GPS outage would be far more widespread. Aviation relies on GPS for navigation. The US Federal Aviation Administration's NextGen air traffic control system is GPS-based. Shipping relies on GPS for routing. The automatic identification system (AIS) used by ships to avoid collisions uses GPS. Agriculture uses GPS-guided tractors for precision farming. Disaster response uses satellite communications and earth observation. A conflict that destroys or disables satellite networks would cause immediate and severe economic damage to every country, regardless of whether that country is a party to the conflict.

Under the laws of armed conflict, the use of ASAT weapons against GPS satellites would raise serious issues under the principle of distinction. GPS satellites are dual-use objects. They serve both military and civilian functions. Under Article 52(2) of Additional Protocol I, a military objective is an object which by its nature, location, purpose or use makes an effective contribution to military action and whose total or partial destruction, capture or neutralization, in the circumstances ruling at the time, offers a definite military advantage. A GPS satellite clearly makes an effective contribution to military action. It provides navigation and timing for military

forces. However, the same satellite also makes an effective contribution to civilian life. The question is whether attacking a GPS satellite offers a "definite military advantage" that outweighs the civilian harm. Given the catastrophic civilian consequences of losing GPS, the use of kinetic ASAT weapons against GPS satellites would likely violate the proportionality rule in Article 51(5)(b) of Additional Protocol I, which prohibits attacks that may be expected to cause incidental loss of civilian life, injury to civilians, damage to civilian objects, or a combination thereof, which would be excessive in relation to the concrete and direct military advantage anticipated.

The Forced Blindness Problem

The current space law framework suffers from “*forced blindness*”. After non Kinetic ASAT attacks like cyberattack, jamming, etc. victim states often cannot gather evidence to prove responsibility. Since the Liability Convention required *proof of fault*, only major space powers with advanced tracking systems can reliably attribute attacks, while most developing states must rely on others for information, creating a major evidentiary and justice barrier.

The Space Insurance Market Asymmetry

The space insurance market is another structural inequity. Satellites are insured against launch failure, in-orbit failure, and third-party liability. The premiums are determined by risk. A state that launches a satellite into a debris-filled orbit pays higher premiums. However, the states that create debris (through ASAT tests or negligent operations) do not pay for the increased risk they impose on others. This is an externality. The Liability Convention does not address this. A Western mega-constellation operator can self-insure because it has the financial resources. A developing nation launching its first satellite cannot self-insure and must pay commercial premiums that reflect the debris risk created by others. This is a regressive transfer of wealth from the Global South to the spacefaring powers. If an ASAT attack creates a debris cloud, the insurance market experiences a "systemic premium shock." All satellites in the affected orbit face increased risk. Premiums increase for everyone. The developing nation that cannot self-insure faces a disproportionate burden. It may be priced out of the market entirely. The Liability Convention does not address this. The state that created the debris is not liable for the increased insurance premiums of other states. The concept of "fault" under the Liability Convention does not include economic harm. Only physical damage to a space object triggers liability. A systemic premium shock is not physical damage.

8. INTERNATIONAL INSTRUMENTS AND SOFT LAW PERTAINING TO THE AGENDA

a) The Outer Space Treaty of 1967 and the ‘Province of All Mankind’ Principle.

OST is the cornerstone. Its Article I is particularly important for developing nations: "The exploration and use of outer space, including the moon and other celestial bodies, shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development, and shall be the province of all mankind." This provision is often cited by developing nations to argue against the "enclosure" of the orbital commons by spacefaring powers. However, the provision is not self-executing. There is no mechanism to enforce it. The question of whether a state that launches a satellite has an obligation to share the data or the benefits of that satellite with other states has never been litigated. The UN General Assembly has repeatedly reaffirmed the principle in resolutions, but it has not created an enforcement mechanism.

b) The Liability Convention of 1972: Fault, Forensics, and the Burden of Proof

The Liability Convention is the only mechanism for financial compensation for space-related damage. However, it has never been used to resolve a dispute involving ASAT weapons. The

convention requires the victim to prove "fault" for damage in space. The standard of proof is not specified. The convention's Claims Commission has never been convened. The convention does not address the allocation of liability for debris created by a third state. If State A destroys its own satellite and the debris damages State B's satellite, is State A liable? The convention's text is ambiguous. The convention defines "space object" to include "component parts of a space object as well as its launch vehicle and parts thereof." Debris is a component part. However, State A could argue that the debris is no longer a "space object" because it is not functioning. This is untested.

c) The Registration Convention of 1975: Transparency Gaps

The Registration Convention requires states to register space objects with the UN, providing information about orbit and function. However, states are not required to declare whether a satellite carries a weapon. The convention does not require pre-launch notification of RPOs. The convention does not require states to disclose changes in orbit or function. The convention does not require states to disclose the ownership or operation of private satellites. SpaceX's Starlink satellites are registered by the United States, but the registration does not disclose the military uses of the constellation. The convention's gaps are significant.

d) The Partial Test Ban Treaty of 1963 and Nuclear ASATs

The PTBT prohibits nuclear tests in space. However, it does not prohibit the deployment of nuclear-armed ASAT weapons. It does not prohibit testing of non-nuclear components. It does

not provide for verification or enforcement. The only nuclear test in space, Starfish Prime, took place before the treaty entered into force. The PTBT's verification mechanism relies on national technical means (seismic monitoring, atmospheric sampling). For a space test, the PTBT's verification is limited. A state could test a nuclear device in deep space, far from Earth, and the PTBT's verification system might not detect it. The PTBT is therefore an incomplete barrier.

e) The Anti-Ballistic Missile (ABM) Treaty of 1972

The ABM Treaty, although no longer in force, established important precedents for space arms control. It limited the deployment of systems designed to shoot down ballistic missiles. It included provisions for verification, including national technical means (satellites). It prohibited the development, testing, and deployment of space-based ABM systems. The ABM Treaty demonstrated that space-based weapons are not inevitable. They can be prohibited by treaty. The ABM Treaty also demonstrated that verification is possible. The United States and the Soviet Union exchanged data on their ABM systems. They permitted on-site inspections. They used national technical means (satellites) to monitor compliance. The ABM Treaty regime collapsed because the United States withdrew, not because it was ineffective.

f) The Wassenaar Arrangement and MTCR: Export Controls as Technical Blockades

The Wassenaar Arrangement on Export Controls for Conventional Arms and Dual-Use Goods and Technologies is a multilateral export control regime (not a treaty) with 42 participating

states. Its purpose is to contribute to regional and international security and stability by promoting transparency and greater responsibility in transfers of conventional arms and dual-use goods and technologies, thus preventing destabilizing accumulations. Participating states voluntarily implement export controls on listed items. The Wassenaar Arrangement's control lists include a wide range of space-related technologies: satellite components, propulsion systems, materials, and software.

The Missile Technology Control Regime (MTCR) is a similar multilateral export control regime (not a treaty) with 35 participating states. The MTCR restricts the transfer of missile and drone technology capable of delivering weapons of mass destruction. The MTCR's control lists include rocket and missile technology, including technology for satellite launch vehicles.

Developing nations argue that the Wassenaar Arrangement and the MTCR function as a cartel, restricting access to space technology while claiming to be non-proliferation measures. The same technology that can be used to build a ballistic missile is also used to build a satellite launch vehicle. A nation that wants to develop an independent space launch capability finds that it cannot legally purchase the necessary components from any Wassenaar or MTCR member state. This creates a structural barrier to entry. The only way to acquire launch capability is to develop it indigenously, which is expensive and time-consuming, or to purchase launch services from a spacefaring power, which creates dependency. The UN General Assembly has repeatedly called for the transfer of space technology to developing nations, but the Wassenaar Arrangement and MTCR are outside the UN framework and are not bound by UN resolutions.

g) The Political Declaration on Responsible Behavior in Outer Space

(2023 – Present)

This is a non-binding political declaration initiated by the United States and currently endorsed by over 40 states. It commits endorsing states to refrain from destructive direct-ascent ASAT testing and to develop further norms of responsible behavior. The declaration is not a treaty. It does not have an enforcement mechanism. It does not include Russia, China, or India. The declaration is significant because it represents the first time a large group of states has committed to a ban on debris-creating ASAT tests. However, its practical effect is limited because the states that have conducted ASAT tests (China, Russia, India) have not endorsed it.

h) UNGA Resolutions on ‘No First Placement of Weapons in Outer Space’ (NFP)

Russia and China annually propose a resolution in the First Committee calling for a political commitment by all states to not be the first to place weapons in outer space. The United States and its allies vote against it, arguing that it is unverifiable and that it does not ban terrestrial ASATs. The NFP resolution has been adopted by the General Assembly every year since 2014, with a majority of states voting in favor. The voting pattern reveals the geopolitical divide: Russia, China, and most developing nations vote in favor; the United States, European allies, and some other states vote against. The NFP resolution is an example of soft law. It is not binding, but it represents the view of the majority of UN member states.

i) The Draft Treaty on the Prevention of the Placement of Weapons in Outer Space (PPWT)

Russia and China jointly submitted a draft treaty to the Conference on Disarmament in 2008 and a revised version in 2014. The PPWT would ban the placement of any weapon in outer space. The United States and NATO rejected it for several reasons. First, it did not ban terrestrial ASATs. A state could destroy satellites from Earth without violating the treaty. Second, it had no verification mechanism. There were no provisions for on-orbit inspections, mandatory data sharing, or challenge inspections. Third, it allowed space-based missile defense. The United States wanted to preserve the option of space-based interceptors. Russia and China wanted to ban them. The PPWT is currently stalled. No state has ratified it.

j) The International Code of Conduct for Outer Space Activities

The European Union led an effort from 2012 to 2015 to create a voluntary, politically binding code of conduct. The code would establish transparency and confidence-building measures (TCBMs), such as pre-launch notifications and information sharing on orbital maneuvers. The effort stalled due to opposition from Russia and China. Russia and China argued that the code was an attempt to bypass the Conference on Disarmament and that it did not address the core issue of weapons placement. The EU has not formally withdrawn the draft code, but it is not actively negotiating it.

k) UN Group of Governmental Experts (GGE) and Open-Ended Working Group (OEWG) Reports

The UN has convened multiple GGEs (small groups of experts appointed by the Secretary-General) and OEWGs (open to all member states) to study space security. The 2013 GGE report (A/68/189) produced a consensus report with recommendations on TCBMs. The report recommended pre-launch notifications, information sharing on orbital maneuvers, and the establishment of a UN database on TCBMs. Later GGEs (2018-2019) failed to reach consensus due to geopolitical tensions. A new OEWG on space security is ongoing (2024-2028). Its mandate is to discuss threats to space systems and possible norms, rules, and principles for responsible behavior. The OEWG is the primary UN forum for the present agenda.

l) The 21 Guidelines for the Long-Term Sustainability (LTS) of Outer Space Activities

These guidelines were developed by COPUOS and adopted by the General Assembly in 2019. They are non-binding technical guidelines for mitigating space debris, sharing information, and coordinating space activities. They do not address weapons, but they establish the principle that states have a duty to act responsibly in space. Guideline A.6 states that states should "take measures to minimize the risks of collision and to avoid the creation of debris." Guideline B.1 states that states should "share information on space weather and space debris." The LTS guidelines are a foundation for further work.

m) The Artemis Accords

The Artemis Accords are a non-UN set of principles for lunar exploration, led by the United States and signed by over 30 states. The Accords include a commitment to "non-interference" (not attacking other's lunar assets), to the registration of space objects, and to the extraction and use of space resources. The Accords are controversial because they are not negotiated within the UN framework and are seen by some states as an attempt to bypass COPUOS and the Outer Space Treaty's common heritage principles. Russia and China have not signed the Accords. The Accords are an example of a "coalition of the willing" approach to space governance.

9. INTERESTING AND UNIQUE ASPECTS OF THE AGENDA

Representatives are encouraged to move beyond general statements and engage with the following interesting and analytically unique aspects of the agenda.

a) The 'Space Basing' Loophole

The Outer Space Treaty does not ban space-based conventional weapons. A state could legally place a missile interceptor in orbit today. The ABM Treaty, although defunct, explicitly prohibited space-based ABM systems. The US withdrawal from the ABM Treaty in 2002 removed that prohibition. The debate over space-based missile defense is therefore alive and well. Some states argue that space-based interceptors are defensive and should be permitted.

Others argue that they are indistinguishable from offensive weapons and would destabilize the nuclear balance. The ABM Treaty's prohibition on space-based interceptors was based on the recognition that such systems would create a "first-strike" incentive. If one state has a space-based interceptor, it could shoot down the other state's retaliatory missiles, creating a first-strike advantage. The other state would then be incentivized to launch first. This is the same logic that applies to ASAT weapons. The ABM Treaty's ghost haunts the present agenda.

b) The Attribution Problem and Forced Blindness

Attributing a non-kinetic ASAT attack is extraordinarily difficult. A laser dazzler can be a ground-based system that looks like an astronomical telescope. A cyber-attack can be routed through servers in three different countries. A jamming signal can be transmitted from a ship that is legally in international waters. The victim state may not have the technical capability to trace the attack to its source. This is "forced blindness." Even if the victim state knows who attacked it, it may not be able to prove it to the standard required by the Liability Convention. The Liability Convention's standard of proof is "fault." For a kinetic attack, fault can be proven through trajectory analysis. For a non-kinetic attack, there is no trajectory. There is only the effect. The victim state must prove that the attacking state intended to cause the damage or was negligent. Without access to the attacking state's internal records, this is impossible.

c) The Verification Dilemma and the Impossibility of On-Orbit Inspections

The Outer Space Treaty provides for on-orbit inspections in Article XII, but only on celestial bodies. There is no provision for inspecting a satellite in orbit. A state cannot demand to inspect another state's satellite to verify that it does not contain a weapon. The only way to be certain would be to destroy the satellite, which would be an act of war. The verification dilemma is therefore acute. Verification relies on national technical means (satellites, ground-based sensors) and on transparency and confidence-building measures (data sharing, pre-launch notifications). National technical means are limited. A satellite's external appearance does not reveal its internal components. A state can hide a laser or a cyber-weapon inside a satellite. The only way to detect it would be to monitor its emissions, but a weapon could be designed to remain dormant until used. The verification dilemma is a core challenge for any legal barrier.

d) The Enclosure of the Orbital Commons: Asymmetric Legal Frameworks and Privatized Warfare

The orbital commons is being enclosed not by formal appropriation (which is banned by Article II of the OST) but by effective control. The states and companies that have the technology to launch and operate satellites control access to the commons. The Wassenaar Arrangement and MTCR create a legal blockade. The Registration Convention provides information asymmetries. The Liability Convention's fault standard benefits the powerful. Privatized warfare, exemplified by SpaceX's Starlink providing military communications to Ukraine, blurs the line between state and commercial actors. Starlink is a commercial system, but it is used for military purposes. If Russia attacks a Starlink satellite, is it attacking the United States (the launching state) or SpaceX (a private company)? Under the OST, the launching state is internationally responsible for all

activities of its nationals, including private companies. However, the OST was written when space activities were conducted exclusively by government agencies. The commercial revolution has outpaced the legal framework. A private company can decide, without state direction, to provide services to one side in an armed conflict. The state of registry may have no control over that decision. The concept of "authorization and continuing supervision" under Article VI of the OST is vague and has never been enforced against a private company.

e) ITAR, MTCR, and the Technical Blockade: Export Controls as a Barrier to Access

The US International Traffic in Arms Regulations (ITAR) and the MTCR restrict the transfer of space technology. Any satellite or launch vehicle that contains US-origin components, no matter how small, is subject to ITAR. This gives the United States effective veto power over many space projects. A state that wants to launch a satellite may find that it cannot purchase a launch vehicle from any commercial provider because the provider's license requires US approval. The MTCR restricts the transfer of rocket technology. A state that wants to develop its own launch vehicle may find that it cannot legally purchase the necessary components from any MTCR member state. Developing nations argue that this violates Article I of the OST, which declares that space shall be the province of all mankind. If a nation cannot acquire the technology to launch its own satellites, it is forever dependent on the spacefaring powers.

f) Non-Kinetic Interference, Directed-Energy Blinding, and Unannounced Rendezvous and Proximity Operations (RPOs) as Deniable Aggression

Non-kinetic interference (jamming, spoofing, dazzling) is the grey zone warfare of space. It is aggressive but deniable. A state can disrupt another state's satellite services without crossing the threshold of an "armed attack" under Article 51. The victim state cannot respond with force, only with countermeasures. This asymmetry benefits the attacker. Unannounced RPOs are particularly dangerous. When a satellite approaches another without prior notification, the target state cannot know whether it is an inspector, a servicing vehicle, or a weapon. The absence of rules for RPOs creates a constant risk of miscalculation and escalation. The target state may assume the worst and respond preemptively, turning a peaceful inspection into an armed conflict. The international community has no agreed rules for RPOs: no mandatory notification, no maximum approach distances, no prohibited maneuvers, no deconfliction channel.

g) The Financial Weaponization of the Space Frontier: Insurance Asymmetries and Externalized Risk

The space insurance market is another structural inequity. Satellites are insured against launch failure, in-orbit failure, and third-party liability. The premiums are determined by risk. A state that launches a satellite into a debris-filled orbit pays higher premiums. However, the states that create debris (through ASAT tests or negligent operations) do not pay for the increased risk they impose on others. This is an externality. The Liability Convention does not address this. A Western mega-constellation operator, such as SpaceX, can self-insure because it has the financial

resources. A developing nation launching its first satellite cannot self-insure and must pay commercial premiums that reflect the debris risk created by others. This is a regressive transfer of wealth from the Global South to the spacefaring powers.

h) The ‘Terrestrial Targeting’ Blind Spot and Ground Segment Vulnerability

As discussed above, the Outer Space Treaty does not protect ground stations. A state could destroy another state's satellite ground station with a cruise missile, and the only applicable law would be the UN Charter (prohibition on the use of force) and IHL (targeting rules). The specific protections for space objects do not apply. The Liability Convention's protection applies to damage caused by a space object, not damage caused to a space object. An attack on a ground station is an attack on terrestrial infrastructure. This gap is significant because ground stations are more vulnerable than satellites. A ground station is a fixed target on Earth. It cannot maneuver. It cannot hide. It is a building with antennas. A single cruise missile can destroy it. The attacker would need to justify the attack under IHL (the ground station must be a military objective) and the UN Charter (the attack must be in self-defence or authorized by the Security Council). But there is no space-specific prohibition.

10. CASE STUDIES OF SPACE WEAPONIZATION, REGULATION ATTEMPTS, AND STRUCTURAL INEQUITIES

Examining specific events and structural conditions provides valuable insights into the legal, political, and economic dimensions of space security. The following case studies illustrate both the dangers of weaponization and the systemic inequities that must be addressed.

Successful and Failed ASAT Tests

1. The 2019 Indian ASAT Test: Responsible Testing or Dangerous Precedent?

On March 27, 2019, India destroyed one of its own satellites (Microsat-R) in low Earth orbit using a direct-ascent ASAT missile (Prithvi Defence Vehicle). India chose a very low orbit (approximately 300 kilometers) so that debris would decay quickly due to atmospheric drag. India claimed it was acting responsibly and that the debris would clear within weeks. NASA administrator Jim Bridenstine noted that the test created 400 pieces of trackable debris and that even a small cloud of debris poses a threat to the International Space Station. The ISS orbits at approximately 400 kilometers, above the altitude of the test. The debris cloud was initially at a lower altitude, but atmospheric drag varies, and some debris could have been pushed to higher altitudes.

The international response was mixed. The United States expressed concern but did not formally condemn the test. Russia called the test concerning. China expressed opposition. India defended

the test as a legitimate national security measure. The test raised important questions about the definition of "responsible" ASAT testing. Is a low-orbit test acceptable because debris decays quickly? Or is any debris-creating test unacceptable because of the risk of collision? The 2023 US-led political declaration on responsible behavior calls for a complete ban on destructive direct-ascent ASAT tests, without exception for low-orbit tests. India has not signed the declaration.

Failed Treaty Negotiations

2. The Failure of the PPWT Negotiations (2008-2014): Lessons from a Collapsed Treaty

The PPWT was proposed by Russia and China in 2008 and revised in 2014. The draft treaty (2008) had 13 articles. It would have banned the placement of any weapon in outer space. It would have banned the use of force against space objects. It did not ban terrestrial ASATs. It did not include a verification mechanism. It allowed space-based missile defense. The United States and NATO rejected it for three reasons. First, it did not ban terrestrial ASATs. A state could destroy satellites from Earth without violating the treaty. Second, it had no verification mechanism. There were no provisions for on-orbit inspections, mandatory data sharing, or challenge inspections. Third, it allowed space-based missile defense. The United States wanted to preserve the option of space-based interceptors. Russia and China wanted to ban them.

The revised draft (2014) addressed some of these concerns. It included a provision requiring states to declare their space objects. It included a provision for a consultative mechanism. It did

not include on-orbit inspections. It still allowed space-based missile defense. The revised draft was also rejected. The PPWT is currently stalled. No state has ratified it. The failure of the PPWT demonstrates that a treaty that does not address terrestrial ASATs and does not include verification is unacceptable to the major spacefaring powers. Any future treaty must be comprehensive (space-based and terrestrial weapons) and verifiable.

Structural Inequities

3. Case Study: The Amazon Rainforest – Monitoring, Data Denial, and Environmental Warfare

The Amazon rainforest is monitored by satellites. The Brazilian National Institute for Space Research (INPE) uses satellite data to track deforestation. The data is publicly available and is used by environmental NGOs, governments, and researchers. Deforestation monitoring is a form of environmental protection. It is also a political tool. When INPE published data showing increased deforestation under President Bolsonaro, the Brazilian government accused INPE of lying and fired its director. The data was not disputed; the interpretation was.

Satellite data can be denied. A state that controls a satellite can choose not to collect data over a certain area, or can refuse to share the data. If the Amazon were to be targeted by an ASAT weapon, the consequences for global climate would be catastrophic. The Amazon absorbs carbon dioxide and produces oxygen. A kinetic ASAT attack on a satellite over the Amazon would create debris that could damage other satellites. But a more subtle threat is data denial. If a state with a satellite over the Amazon decides to stop sharing data, the international community loses

visibility into deforestation. Data denial is not a use of force. It is not prohibited by any treaty.

The Outer Space Treaty's "due regard" obligation could be interpreted to require data sharing for environmental monitoring, but this is untested.

4. Case Study: The Space Insurance Market – Self-Insured Western Constellations and Systemic Premium Shocks for the Global South

The space insurance market is a multi-billion dollar industry. Satellites are insured against launch failure, in-orbit failure, and third-party liability. Premiums are determined by risk. A satellite launched into a debris-filled orbit pays higher premiums. A satellite with a proven track record pays lower premiums. Western mega-constellations (Starlink, OneWeb) are self-insured. SpaceX does not purchase commercial insurance for Starlink satellites. The company self-insures because it has the financial resources. A developing nation launching its first satellite cannot self-insure. It must purchase commercial insurance. The premiums reflect the debris risk created by all space operators, including Western mega-constellations. This is a regressive transfer of wealth. The developing nation pays for the risk created by others.

If an ASAT attack creates a debris cloud, the insurance market experiences a "systemic premium shock." All satellites in the affected orbit face increased risk. Premiums increase for everyone. The developing nation that cannot self-insure faces a disproportionate burden. It may be priced out of the market entirely. The Liability Convention does not address this. The state that created the debris is not liable for the increased insurance premiums of other states. The concept of

"fault" under the Liability Convention does not include economic harm. Only physical damage to a space object triggers liability. A systemic premium shock is not physical damage.

5. Case Study: The Starfish Prime Test (1962) – The Only Nuclear Detonation in Space

On 9 July 1962, the United States detonated a 1.4-megaton nuclear weapon 400 km above the Pacific during Operation Fishbowl. The explosion generated an EMP that destroyed 9 of 25 satellites in orbit and created a radiation belt that damaged additional satellites for years. The test showed that nuclear ASAT weapons are inherently indiscriminate, affecting civilian and military satellites alike.

Although the Partial Test Ban Treaty prohibits nuclear testing in space, it does not explicitly ban the deployment of nuclear-armed ASATs. While such deployment would violate the Outer Space Treaty, verification remains difficult. Starfish Prime remains a stark warning of the catastrophic consequences of nuclear weapons in space and highlights the need to address nuclear-armed ASAT threats alongside conventional ASATs.

11. ROLE OF OTHER UNITED NATIONS BODIES AND EXTERNAL ACTORS

a) The Conference on Disarmament (CD)

The CD is the sole multilateral disarmament negotiating forum. It has 65 member states. It operates on the basis of consensus. The CD has been deadlocked for over two decades. The deadlock is due to the inability to agree on a Programme of Work. The CD has not held substantive negotiations on space weapons since the 1990s. The CD's deadlock is a major roadblock to a legally binding treaty.

b) The UN Office for Outer Space Affairs (UNOOSA)

UNOOSA is the secretariat for COPUOS. It is located in Vienna. UNOOSA provides technical assistance to states on space law and policy. It maintains the UN Register of Space Objects. UNOOSA does not have a disarmament mandate. Its role is to promote peaceful uses.

c) The International Telecommunications Union (ITU)

The ITU is a UN specialized agency. It regulates radio spectrum and orbital slots. The ITU has no mandate for weapons. The ITU's Constitution prohibits "harmful interference" with radio

services. Military jamming could be considered harmful interference, but the ITU has no enforcement mechanism against states.

d) The First Committee (Disarmament and International Security)

The First Committee deals with nuclear weapons, conventional arms, and other security threats. The First Committee adopts annual resolutions on the prevention of an arms race in outer space. The First Committee and the Fourth Committee have overlapping mandates on space security. Coordination is necessary to avoid duplication.

e) The Security Council

The Security Council has primary responsibility for international peace and security. An attack on a satellite network could be considered a threat to peace under Article 39 of the UN Charter. The Security Council could issue binding resolutions on space weapons. However, the P5 (US, Russia, China, UK, France) have veto power. Russia and China have vetoed space-related resolutions in the past.

f) The International Court of Justice (ICJ)

The ICJ has jurisdiction over disputes between states. A state could bring a case against another state for violating the Outer Space Treaty or the Liability Convention. The ICJ could issue an

advisory opinion on the legality of space weapons. The UN General Assembly could request an advisory opinion. An advisory opinion would clarify the law but would not be binding.

g) Non-UN Actors: European Space Agency (ESA), NATO, Commercial Satellite Operators

ESA is an intergovernmental organization with 22 member states. ESA has a space security program. NATO has declared space an operational domain. NATO members have invoked Article 5 (collective self-defence) for attacks on space assets? Not yet. Commercial operators (SpaceX, OneWeb, etc.) have their own security policies. A treaty that binds states must also address commercial operators.

12. RESEARCH QUESTIONS

1. What Constitutes a "Weapon in Orbit"?
2. Should Conventional Orbital Weapons Be Banned?
3. Defining "Peaceful Use of Outer Space"
4. Should Kinetic ASAT Tests Be Universally Prohibited OR Can They Be Regulated ?
5. Should Cyberattacks Against Satellites Be Treated as Acts of War?
6. Is Space Debris a Security Threat?

7. To what extent do national export-control regimes, technology-transfer restrictions, and launch-access limitations affect the principle that outer space shall be the province of all humankind, and how can equitable access to space technologies be promoted?
8. What reforms are necessary to ensure that commercial expansion into Outer Space- including mega constellations, private launch providers, and emerging space security industries- does not undermine equitable access, sustainability and international peace and security?

13. CONCLUDING NOTE

As a United Nations General Assembly SPECPOL, the committee is expected to focus on ***consensus-oriented deliberation*** and ***collaborative problem-solving***, where representatives should recognize that the General Assembly does *not possess enforcement authority* comparable to that of the Security Council. Instead, the Assembly's strength lies in its ability to articulate *political commitments, develop normative frameworks, and encourage coordinated action* among Member States.

Consequently, resolutions produced within this committee should aim to:

- 1) *Provide clear recommendations to Member States and relevant United Nations (UN) bodies*
- 2) *Encourage institutional coordination across the UN system*
- 3) *Establish reporting mechanisms or requests to the Secretary-General*
- 4) *Promote capacity-building initiatives for developing states*
- 5) *Encourage confidence-building measures and transparency frameworks*

Note: Representatives are encouraged to draft resolutions that are realistic, implementable, and consistent with the mandate of the General Assembly.

Strong resolutions typically contain:

- 1) Clearly defined operative clauses*
- 2) References to existing international frameworks*
- 3) Institutional proposals that fall within the General Assembly's competence*
- 4) Balanced consideration of political feasibility and normative ambition*

Epilogue: Ultimately, the goal of the committee is not merely to produce declarations but to craft policy recommendations that meaningfully contribute to strengthening international cooperation in conflict prevention and technological governance.

